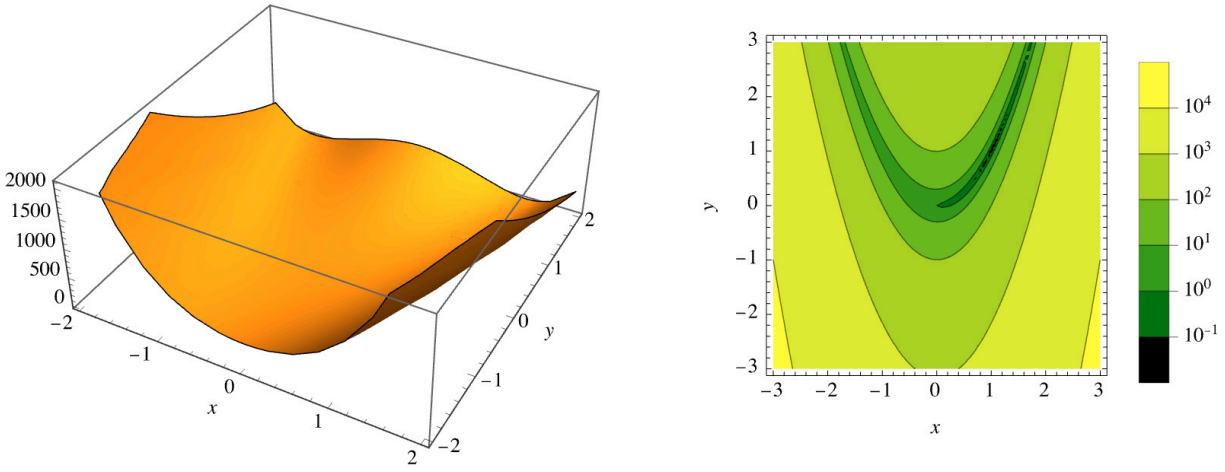
Rosenbrock Function

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The Rosenbrock function, also known as Rosenbrock's banana function or Rosenbrock's valley, is the function defined by

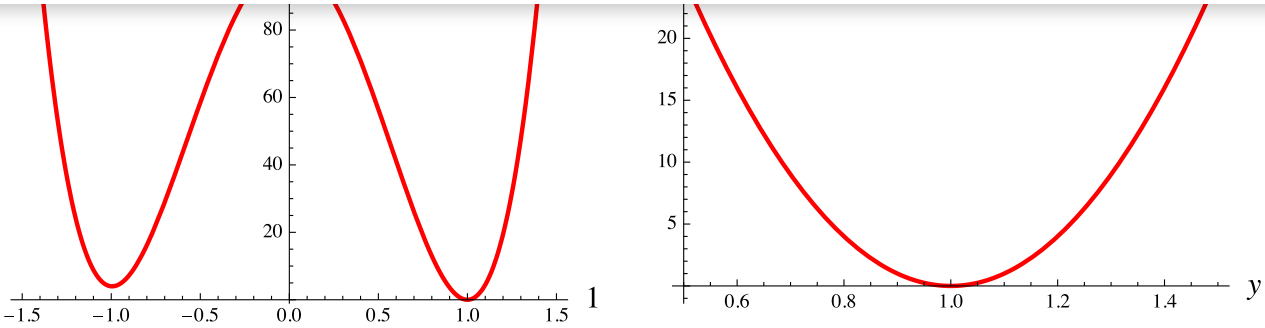
$$f(x, y) = (a - x)^2 + b\left(y - x^2\right)^2$$

that is often used as a test problem for optimization algorithms due to the existence of a global minimum of 0 at the point $\left(a, a^2\right)$ which is difficult to find.



The parameters are most commonly taken as $a = 1$ and $b = 100$ (though $b = 105$ is sometimes also used; Germundsson 2000), as illustrated above in the $x\,y$ plane.

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The function with $a = 1$ and $b = 100$ is plotted above for $y = 1$ and $x = 1$.

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REFERENCES

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Rosenbrock, H. H. "An Automatic Method for Finding the Greatest or Least Value of a Function." *Computer J.* **3**, 175-184, 1960.

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